

Full Length Article

Surface dynamics of solid-lubricated PA66 with glass fibers under high load

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ABSTRACT

Polyamide 66 without additives exhibits mechanical properties that are too weak for many structural applications. Young's modulus and yield stress can be improved by incorporating stiff glass fibers; however, this simultaneously increases the friction coefficient—an undesirable effect in mechanical components such as bearings or gear wheels. Solid lubricants, especially PTFE, are widely used to reduce and stabilize friction and wear. Due to environmental concerns, PTFE is under legislative pressure and is expected to be replaced. This systematic study includes ball-on-disk experiments with Hertzian pressure close to yield stress, using different solid lubricants (e.g. PTFE, graphite, molybdenum disulfide, and a further PTFE-free lubricant inferred to be polymer-based). While friction coefficients are recorded, the interfacial dynamics during sliding are observed using a microscope camera. Synchronizing the camera's frame rate with the rotational speed of the sample enables capturing images of precisely the same surface location over time. The resulting pictures reveal the transport of wear particles, their agglomeration into a thin layer and its removal. These surface dynamics differ significantly depending on the solid lubricant employed. The optical observations can be correlated with the coefficient of friction. The friction coefficient is governed by converging running-in behavior and a quasi-stationary state. This quasi-stationary state consistently falls into one of two characteristic levels—a lower or higher plateau—depending on the lubricant applied. A comprehensive understanding of the friction mechanisms thus offers a basis for identifying appropriate friction modifiers.

1. Introduction

Polyamide 66 (PA66) is one of the most widely used engineering thermoplastics in tribological applications due to its favorable balance of mechanical strength, thermal stability, and chemical resistance. Typical application fields include sliding bearings, gears, and structural components exposed to repeated contact loading and relative motion [1]. However, unfilled PA66 exhibits comparatively low stiffness and yield strength, which limits its use in highly loaded mechanical systems [2].

To overcome these limitations, the incorporation of glass fibers is a well-established strategy. Glass fiber reinforced PA66 shows a substantial increase in Young's modulus, yield stress, and dimensional stability, enabling its application in load-bearing tribological contacts [3]. At the same time, this reinforcement introduces significant drawbacks from a tribological perspective. The presence of stiff fibers close to the surface promotes higher friction coefficients and accelerates counterbody wear, particularly under dry sliding conditions and high contact pressures [4]. As a result, glass fiber reinforced PA66 often requires additional friction modifiers to meet the demands of tribological performance.

Solid lubricants are commonly employed to reduce friction and wear in polymer-based tribosystems. Among these, polytetrafluoroethylene (PTFE) has proven especially effective due to its low shear strength and its ability to form stable transfer films on metallic counterfaces [2]. PTFE-filled PA66 compounds therefore represent a well-established solution in many industrial applications. Nevertheless, increasing environmental and regulatory pressure on per- and polyfluoroalkyl substances (PFAS), to which PTFE belongs, has triggered an urgent need for alternative solid lubricants with comparable tribological performance but reduced environmental impact [5,6]. Potential substitutes include lamellar solid lubricants such as graphite and molybdenum disulfide (MoS₂), which are known to reduce friction by facilitating interlayer shear. While these materials are widely studied in metallic tribosystems, their effectiveness in glass fiber reinforced polyamide matrices—especially under high contact pressures—remains insufficiently understood [7,8]. In particular, the interaction between glass fibers, polymer matrix, solid lubricant particles, and the metallic counterbody leads to complex surface and subsurface processes that cannot be captured by friction measurements alone.

Most tribological studies on polymer composites focus on macroscopic observables such as the coefficient of friction and the wear

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rate. Although these quantities are essential for material ranking, they provide limited insight into the underlying mechanisms governing friction stabilization, running-in behavior, and long-term performance. Recent studies have emphasized the importance of in situ and operando observation techniques to resolve interfacial processes such as transfer film formation, debris generation, and third-body dynamics [9]. For polymer-based tribosystems under high load, however, such experimental approaches are still scarce.

In highly loaded ball-on-disk contacts, where Hertzian contact pressures approach the yield stress of the polymer composite, frictional behavior is governed by a dynamic balance between material deformation, wear particle generation, and the formation of transient lubricating layers. The running-in phase and the subsequent quasi-stationary friction regime are particularly sensitive to the type of solid lubricant employed [7,8]. A systematic correlation between surface dynamics and friction evolution is therefore essential to identify the dominant friction mechanisms.

The present study addresses this gap by combining conventional ball-on-disk tribometry with synchronized in situ and operando optical surface observation. By matching the frame rate of a microscope camera to the rotational speed of the disk sample, identical surface locations are repeatedly imaged throughout the sliding experiment. This approach enables direct visualization of transient interfacial processes such as wear particle formation, transport, agglomeration into compacted surface layers, and their sudden removal, which cannot be resolved by friction measurements alone. The study provides a systematic comparison of different classes of solid lubricants, including PTFE, lamellar lubricants (graphite and MoS_2), and an inferred polymer-based lubricant, in glass fiber reinforced PA66 under Hertzian contact pressures close to the yield stress of the polymer matrix. The evolution of the coefficient of friction is correlated with lubricant-specific surface dynamics, revealing running-in behavior, quasi-stationary friction plateaus, and abrupt friction spikes associated with interfacial layer instabilities. By establishing a direct link between surface dynamics and friction stabilization mechanisms, this work advances the mechanistic understanding of solid-lubricated glass-fiber reinforced Polyamide 66 (GF-PA66) under severe loading conditions. The results provide a physically grounded basis for the evaluation of PTFE-free friction modifiers and highlight the importance of interfacial dynamics for the design of tribologically optimized polymer composites.

2. State of the art

PA66 is a widely used engineering thermoplastic for tribological applications due to its favorable combination of strength, thermal stability, and chemical resistance. From a materials science perspective, PA66 is a semi-crystalline polymer, consisting of crystalline lamellae embedded in an amorphous matrix. The relative fraction and morphology of these phases govern key mechanical properties such as stiffness, yield stress, and toughness, as well as tribological behavior. The crystalline regions provide load-bearing capability and thermal resistance, whereas the amorphous regions contribute to ductility and energy dissipation. Under tribological loading, particularly at high contact pressures, deformation and wear are therefore governed by the local response of both phases. This heterogeneous microstructure becomes especially relevant in dry sliding contacts where stresses approach the yield stress of the amorphous phase [10].

A critical aspect of PA66 is the strong dependence of its material properties on moisture content. Water acts as a plasticizer by preferentially interacting with the amorphous regions via hydrogen bonding, leading to a reduction in Young's modulus, yield stress, and hardness, while increasing ductility. From a tribological standpoint, increased moisture content can reduce friction by lowering the shear strength of the surface layer, but may also promote higher wear rates under high loads due to enhanced plastic deformation [2].

Glass fiber reinforcement is commonly employed to improve the stiffness and strength of PA66. A typical glass fiber content of 30 wt% significantly increases Young's modulus and reduces creep and thermal deformation. However, glass fibers introduce pronounced tribological drawbacks. The exposure of stiff fibers at the sliding interface increases plowing and micro-abrasive mechanisms, leading to higher friction coefficients and accelerated wear of both the polymer and the metallic counterbody [2,4]. In GF-PA66, the moisture sensitivity of the polymer matrix further complicates tribological behavior, as differential swelling and local softening of the amorphous phase can affect fiber-matrix debonding and debris formation.

Recent studies confirm that the tribological behavior of GF-PA66 is governed not only by the increased stiffness of the composite, but also by fiber exposure, fiber orientation, fiber-matrix adhesion, counterbody material, and environmental conditions. Kim et al. investigated monolithic and glass-fiber reinforced PA66 under humid conditions and showed that water absorption and contact-plateau formation strongly affect friction and wear [11]. Autay et al. systematically studied short glass fiber reinforced PA66 and reported that glass fiber content influences both mechanical properties and dry sliding wear behavior [12]. More recent work by Kunishima et al. demonstrated that, under high contact pressure and grease-lubricated sliding against steel, surface damage involving peeled-off glass fibers and scratching of the PA66 matrix leads to increasing friction, wear, and creep of the composite [13]. The same group further showed that glass fiber diameter, PA66 molecular mass, and fiber-matrix interfacial adhesion affect the wear resistance of GF-PA66 and the abrasive damage imposed on the steel counterbody [14]. These findings underline that GF-PA66 cannot be treated as a simple stiffened polymer in tribological contacts; rather, the sliding response results from the coupled evolution of matrix deformation, fiber exposure, transfer-film formation, and counterbody interaction. This is consistent with the motivation of the present study, where the dynamic surface processes of solid-lubricated GF-PA66 are directly observed under severe loading conditions.

To counteract the unfavorable tribological effects of glass fiber reinforcement, various friction modifiers are employed in polymer composites. These solid additives can be classified into polymeric lubricants (e.g., PTFE, UHMW-PE as a PFAS-free alternative) and lamellar lubricants (e.g., graphite, MoS_2 , h-BN) [2]. PTFE has been widely used in industry in PA66-based tribological compounds due to its low shear strength and its ability to form continuous, stable transfer films on metallic counterfaces. PTFE-filled PA66 and GF-PA66 compounds typically exhibit significantly reduced and stabilized friction coefficients. However, increasing environmental and regulatory concerns regarding fluorinated polymers have intensified research into alternative solid lubricants.

Graphite and MoS_2 are among the most frequently investigated PTFE alternatives. While these additives are effective in many tribosystems, their performance in polymer matrices is less predictable and strongly influenced by environmental conditions, dispersion quality, and contact pressure. Unlike PTFE, lamellar solid lubricants often contribute to friction reduction through third-body mechanisms, involving wear debris, lubricant particles, and fragmented fibers rather than the formation of uniform transfer films [8,15].

Graphite and molybdenum disulfide are classical lamellar solid lubricants, characterized by strong covalent bonding within basal planes and weak interlayer interactions. Their lubricating action is commonly attributed to easy shear between adjacent layers when aligned parallel to the sliding direction [7,16].

For graphite, it is well established that its lubricating effectiveness is strongly dependent on ambient humidity. Adsorbed water molecules or other polar species facilitate interlayer sliding by reducing interfacial adhesion between basal planes. Consequently, graphite exhibits low friction in humid environments, whereas its lubricating performance deteriorates significantly under dry or vacuum conditions [17,18]. This is further influenced by the load level [19]. In polymer composites,

Table 1
List of materials and selected properties in dry and conditioned state.

Compound ID	GF30-A	GF30-PTFE15-A	GF30-Mo2-A	GF30-G5-S	GF30-PO-S
Polymer matrix	PA66	PA66	PA66	PA66	PA66
Glass fiber fraction	30 %	30 %	30 %	30 %	30 %
Lubricant	None	PTFE	MoS ₂	Graphite	Polymer
Density (dry), kg/m ³	1360	1490	1369	1400	1240
Young's modulus (dry/cnd.), GPa	10.0/6.4	9.5/7.0	10.0/7.5	9.8/–	9.0/–
Tensile stress, break 5 mm/min. (dry/cnd.), MPa	180/110	165/125	185/135	156/–	135/–
Tensile strain, break 5 mm/min. (dry/cnd.), %	3.1/5.5	3.2/5.2	3.0/5.8	2.2/–	3.2/–
Sources of materials and material data	[24]	[25]	[26]	[27]	[28]

graphite particles may additionally interact with the polymer matrix and moisture, influencing their mobility and participation in third-body layers. One study varies load and graphite content in PA6, but without consideration of glass fiber content [20].

Molybdenum disulfide also possesses a lamellar crystal structure but differs from graphite in its environmental sensitivity. MoS₂ can provide low friction under dry and vacuum conditions; however, its tribological behavior in humid atmospheres is more complex. Exposure to moisture and oxygen can lead to edge oxidation and the formation of molybdenum oxides, which may increase friction and wear over time [21,22]. In polymer matrices, these effects are superimposed on matrix-controlled phenomena such as plastic deformation, wear particle generation, and lubricant extrusion under high contact pressures. Kumar studied the role of MoS₂ in PA66, but without glass fibers being present [23].

Despite extensive studies on the tribological behavior of PA66, glass fiber reinforced PA66, and lamellar solid lubricants, a comprehensive understanding of how polymer microstructure, moisture content, and lubricant-specific surface dynamics jointly govern friction stabilization under high load remains incomplete. In particular, direct experimental observations linking humidity-dependent surface processes to the evolution of friction are still scarce.

3. Materials and methods

Five market-available material compounds were selected, as shown in Table 1. For each of them, a compound ID has been provided, indicating the ingredients beyond PA66. The fraction of glass fibers is kept identical. The compounds comprise, according to their datasheets, different lubricants: none, PTFE, graphite and MoS₂. The composition of the lubricant used in commercial PTFE-free PA66 compounds “GF30-PO” is not fully disclosed, but likely to be polymer-based (and not lamellar, like graphite and MoS₂).

The designation of GF30-PO as a polymer-lubricated compound is based on an indirect classification approach. The classification is therefore inferred from publicly available datasheet information, macroscopic appearance (e.g. color and homogeneity), and the exclusion of inorganic lamellar lubricants such as graphite or MoS₂, which are typically explicitly declared due to their distinct optical and tribological signatures. Consequently, the term “polymer lubricant” is used throughout this work as a functional descriptor.

The compounds have been molded to disk samples with geometries shown in Figs. 1 and 2.

The mechanical properties of the polymer matrix PA66 depend on the humidity conditions and the fraction of amorphous and crystalline regions, which can be influenced by thermal treatment (tempering). Moreover, the behavior of the solid lubricants can depend on humidity conditions. Consequently, four conditioning states were defined for each material in this study. They comprise dry, humid, and lab-air conditions as well as a thermal tempering process. For humidity conditioning, an exposure time has to be defined. For this, Fick's second law was applied according to [29]. The Fourier number of $Fo = 0.04$ ensures that a minimum of 70% of the stable end state has been reached for the material between the surface and a depth of 200 μm (maximum expected wear depth). The diffusion coefficient $D = 0.32 \cdot 10^{-6} \frac{\text{mm}^2}{\text{s}}$ is

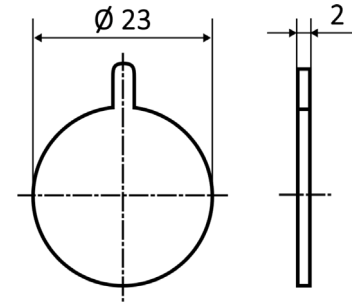


Fig. 1. Geometry of a sample disk, length in millimeters.



Fig. 2. Sample disks of five material compounds.

taken for PA66 at lab temperature, while $s_w = 2 \text{ mm}$ is the thickness of a thin disk conditioned on both sides. An exposure time of six days under constant humidity conditions therefore was defined. Temperature and humidity were monitored by a TP357 (by Thermpro) device.

$$t = \frac{Fo \cdot s_w^2}{D} = \frac{0.04 \cdot (2 \text{ mm})^2}{0.32 \cdot 10^{-6} \frac{\text{mm}^2}{\text{s}}} = 5.8 \text{ days} \quad (1)$$

- **F50:** Lab humidity conditions. The samples are stored at lab conditions for at least 6 days. Lab conditions were not electronically adjusted, but measured: 44...54 %r.H. and 18...22°C
- **F10:** Dry conditions. The samples are stored under dry conditions for at least 6 days within a desiccator with silica gel. The measured conditions in the desiccator were: <10 %r.H. and lab temperature.
- **F75:** Humid conditions. The samples are stored under elevated humidity conditions for at least 6 days within a desiccator in wet air. The air was humidified by the presence of saturated NaCl-solution in the desiccator, but not in direct contact with the samples. The measured conditions in the desiccator were: 75 %r.H. and lab temperature.
- **F50T:** Thermal tempering and subsequent storage for at least 6 days under lab conditions 44...54 %r.H.. The tempering was carried out in a convection oven (UFE700 by Memmert) and by keeping the temperature at 150 °C for two hours. The temperature profile in Fig. 3 demonstrates that this is between the glass transition temperature and the melting temperature to enhance crystallinity. The oven temperature was increased with 20 $\frac{\text{K}}{\text{h}}$ slowly reduced with $-10 \frac{\text{K}}{\text{h}}$. After tempering, the samples again were conditioned for at least 6 days under lab conditions 44...54 %r.H..

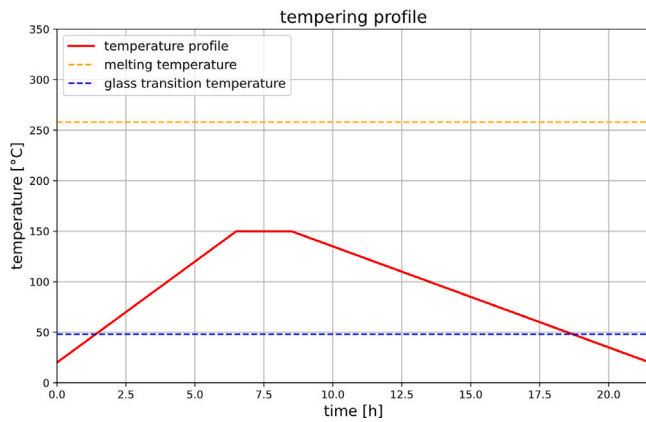


Fig. 3. Tempering temperature profile for F50T.

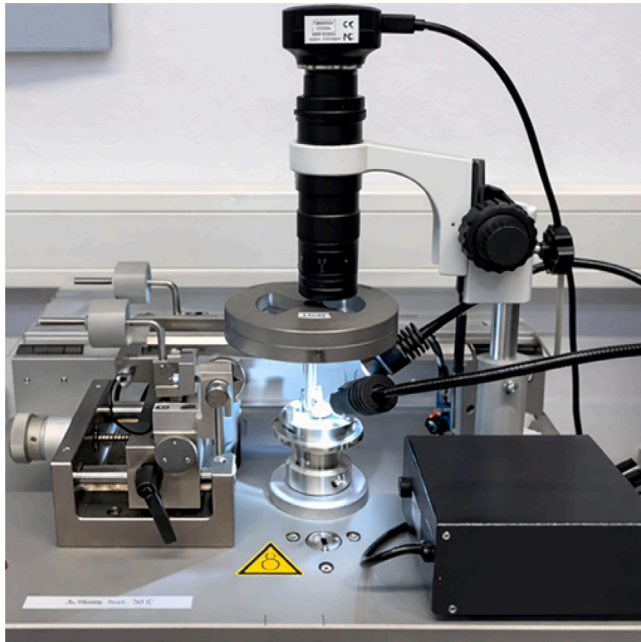


Fig. 4. Tribometer and microscopic camera configuration. To allow an optical path from the camera to the disk sample, two triangular windows have been cut into the dead weights of 10 N each. They are symmetrical to maintain the point of gravity in the center. The radii of the dead weights are adapted to maintain the normal load. Lighting allows for a short camera exposure time to generate sharp pictures while the sample continuously rotates.

The tribological experiments were carried out on a ball-on-disk tribometer (TRB3 by AntonPaar), inspired by ASTM-G99, [30]. The tribometer was equipped with a microscopic camera (MikroCam II 12 MP by Bresser) with a perpendicular view of the free side of the sample disk, Fig. 4. The trigger time for the camera was synchronized with the rotation time of the disk to ensure that a picture of the sliding path is always taken at the identical position. The camera exposure time was sufficiently small to ensure that a sharp picture is taken without stopping the motion of the rotating disk.

Before and after each experiment, the polished steel ball was visually inspected and cleaned with isopropanol. If scratches appeared on the polished surface, the steel ball was replaced. The load parameters are as shown in Table 2. A comparison of the Hertzian contact pressure with the tensile stress at break given in Table 1 shows that the load was chosen high enough to ensure that the material reaches plastic deformation in the middle of the sliding path.

Table 2

Load parameters for the ball-on-disk-experiment.

Normal load	20 N
Ball	100Cr6 bearing steel, diameter 10 mm
Theoretical max. Hertzian contact pressure	193 MPa for $E = 6400$ MPa
Sliding path mid diameter	14 mm
Sliding velocity	70 $\frac{\text{mm}}{\text{s}}$
Sliding distance	800 m
Sliding time	3 h 10 min.

The experimental design included four conditioning states for each of the five materials. In turn, for each of the 20 combinations, 4 individual samples were run in the tribological tests (named “rot_Pr1” to “rot_Pr4”), leading to a total of $5 \cdot 4 \cdot 4 = 80$ tests. The samples were kept in their conditioned environment (desiccator or lab air for at least 6 days) until the experiment (duration: 3 h, 10 min.) started. The test itself was performed under lab conditions.

4. Results

For the unlubricated case (GF30-A-F50, PA66 and glass fibers only) under lab conditions (no humidity or thermal treatment), coefficients of friction are shown in Fig. 5 for four samples. All samples show nearly identical behavior over sliding distance. The running-in in the first 200 m for samples “Pr1” and “Pr2” shows a slight different characteristic compared to the running-in characteristic of samples “Pr3” and “Pr4”. This occurs although all four samples are from the same material and were tested with the same production parameters, conditioning, loads, and lab conditions. Such scattering, which is frequently observed in tribology, can be handled by experimental repetition (here: 4 experiments). Furthermore, this scattering does not impact the run-in state for sliding distances longer than 200 m.

For each sliding distance, the arithmetical average and standard deviation have been calculated, Fig. 6. The coefficient of friction starts at a low level of $\mu \approx 0.05$ and develops to a higher level of $\mu \approx 0.17$ with sliding distance. The change of friction level is accompanied by a phase of higher standard deviation, while for the start and the end of the experiment the standard deviation is small. The change in the coefficient of friction between both levels is attributed to the presence of glass fibers and wear particles in the contact. Initially, both the PA66 matrix and glass fibers are present on the surface and in the contact. Both carry frictional load. However, PA66 under load wears faster than the glass fibers, which have a high wear resistance. Consequently, with increasing sliding distance, glass fibers carry a higher fraction of the load and increase the coefficient of friction.

The last 200 m of the sliding process (600 m to 800 m) are defined as the stable “run-in state”. Here, the coefficient of friction is further evaluated. Results for this “run-in state” are shown in Fig. 7 for all investigated materials and condition states.

Two effects can be distinguished from each other: The applied solid lubricant and the humid or thermal conditioning. The larger impact is provided by the solid lubricant. Here two levels can be recognized. The lowest coefficients of friction can be found for PTFE; the samples with polymer lubricant appear at a similar level. As expected, the system without lubricant provides the highest friction values. Contrary to intuition, the samples with graphite or MoS₂ do not show any friction-reducing effect of the lubricant. A more precise analysis of this will follow in this paper.

The minor impact is conditioning. The less humidity is available (F75-F50-F10), the lower the coefficient of friction is. This can be explained by two effects: First, the presence of water molecules increases adhesion. Second, water molecules in PA66 make the material more ductile, leading to more energy consumption for plastic surface deformation. A similar effect has been observed by Wu et al. for pure PA66 in gear applications [1]. Some sources mention friction reduction

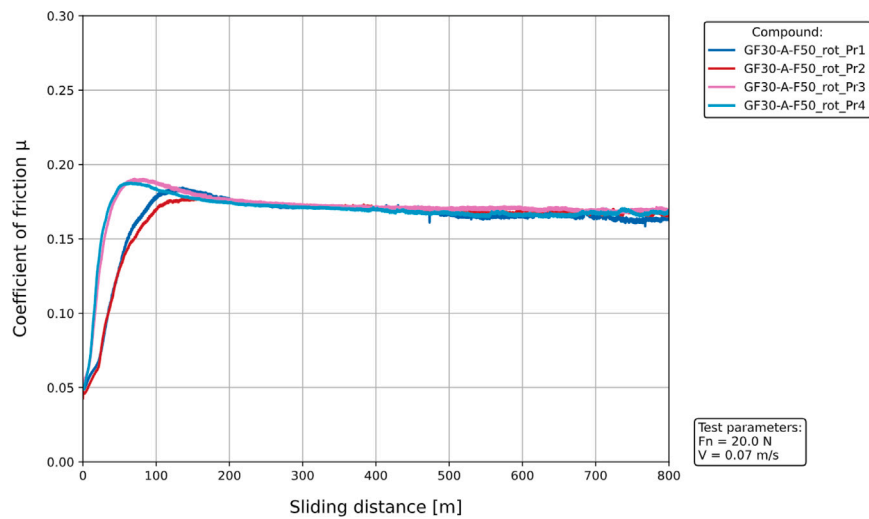


Fig. 5. Coefficients of friction for four samples over sliding distance for GF30-A-F50 (no solid lubricant, no humidity or thermal conditioning).

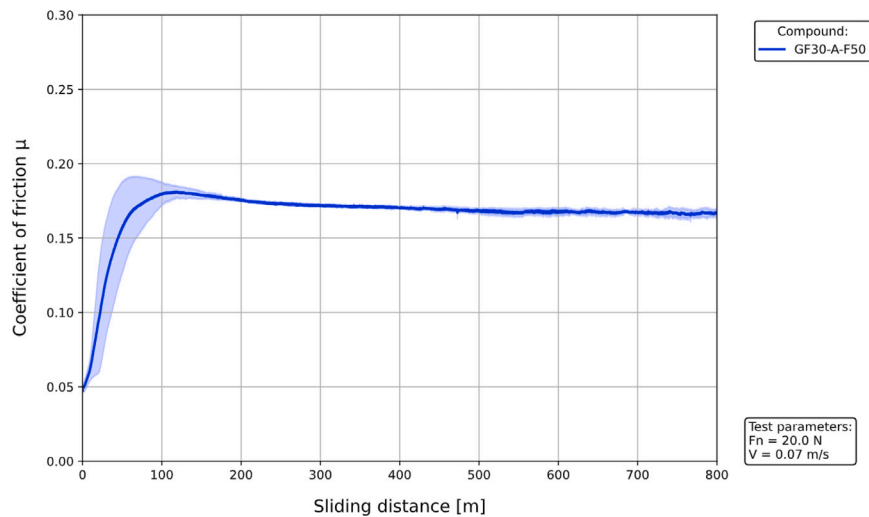


Fig. 6. Averaged coefficients of friction and standard deviation (envelope) over sliding distance for GF30-A-F50 (no solid lubricant, no humidity or thermal conditioning).

of graphite in combination with water molecules [18,19], but this is not found here. The reason may be the influence of Hertzian stress close to the yield stress, the presence of glass fibers, and the humidity sensitivity of PA66. The samples with thermal treatment (F50T) show rather low coefficients of friction. An influence of drying from tempering can be excluded here, as the samples have been stored for at least 6 days in lab humidity conditions after thermal treatment. The reason for the reduced coefficient of friction after tempering is likely increased crystallinity, leading to less ductile behavior of the material.

From the overall data, two scenarios are considered in greater depth: the friction values of the samples with graphite and MoS_2 after the tempering process. Video material generated by in-situ and in-operando imaging applied in this study is published together with this paper on the journal's website.

Fig. 8 shows the coefficients of friction for four samples with graphite over sliding distance. All four samples show the presence of spikes that have a height of $\Delta\mu \approx 0.05$. Between two spikes, the coefficient of friction slowly increases. During the occurrence of one spike, the coefficient of friction falls back to its lower original level.

Fig. 9 shows a section of two typical spike events, including microscopic imaging of the surface change during this event. The sliding path before the spike event is covered by black material. Obviously, black

wear debris from the PA66-matrix and the graphite are compacted and agglomerated here under high frictional load and form a layer on the disk surface. Growth of the surface fraction of this layer over time has been observed while the coefficient of friction is slowly increasing. The authors assume that the increasing load on the layer makes it unstable and leads to an abrupt breakaway of the layer, accompanied by a spike in the coefficient of friction. After the agglomerated layer is removed, the coefficient of friction falls to a minimum and most of the normal load is carried by glass fibers again (white in Fig. 9). The process starts again with a slow increase of the coefficient of friction together with a new buildup of the agglomerated wear particles. This repetitive cycle forms an ongoing limit cycle in the coefficient of friction, which can be explained by the dynamics of the surface layer.

A similar repetitive process can be observed for three of four samples with MoS_2 lubrication in the tempered case, as shown in Fig. 10. Samples “rot_Pr1”, “rot_Pr2” and “rot_Pr3” show abrupt spikes. They are less numerous than those for graphite lubrication, but clearly distinguishable. The process of surface change over one spike event is depicted in Fig. 11. Before a spike event occurs, the surface is covered with a yellow layer, further yellow wear particles can be found on both sides of the sliding path. The yellow layer grows over sliding distance until it is suddenly removed while a spike event occurs. After

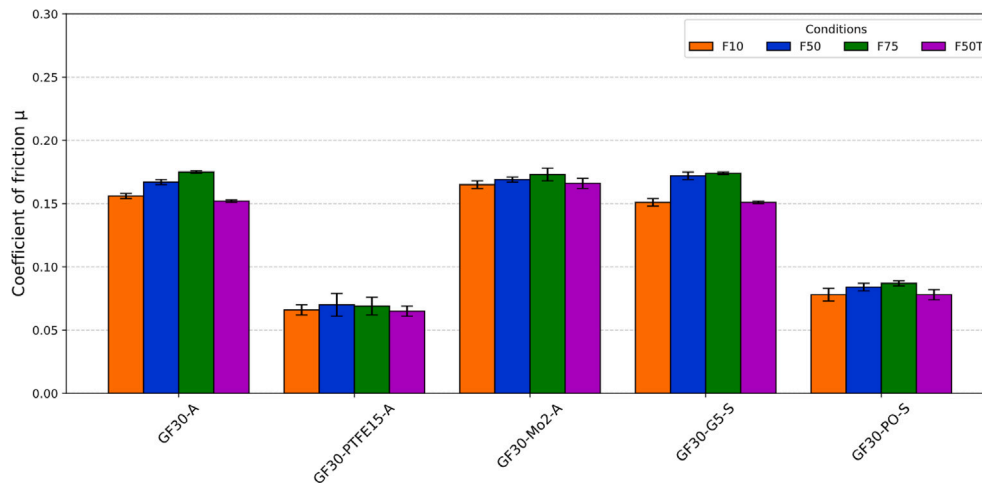


Fig. 7. Coefficient of friction in the run-in-state (600–800 m) for different materials and condition states.

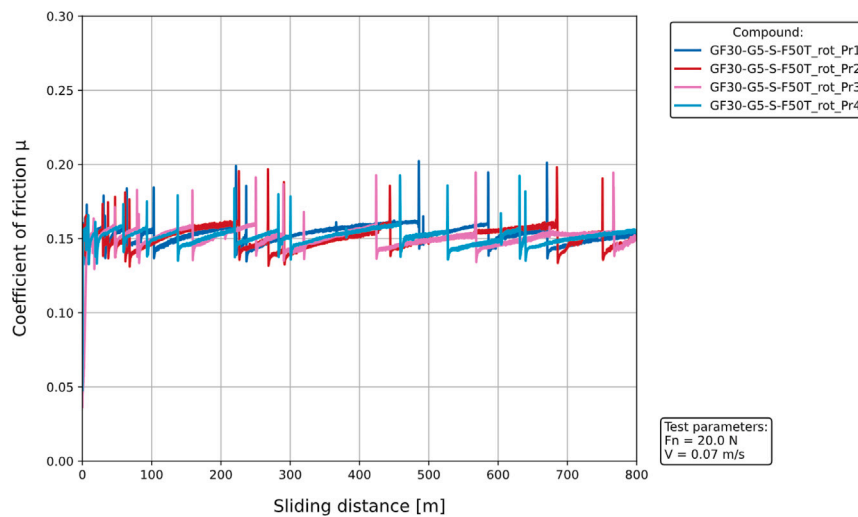


Fig. 8. Coefficient of friction over sliding distance for four samples of GF30-G-S-F50T.

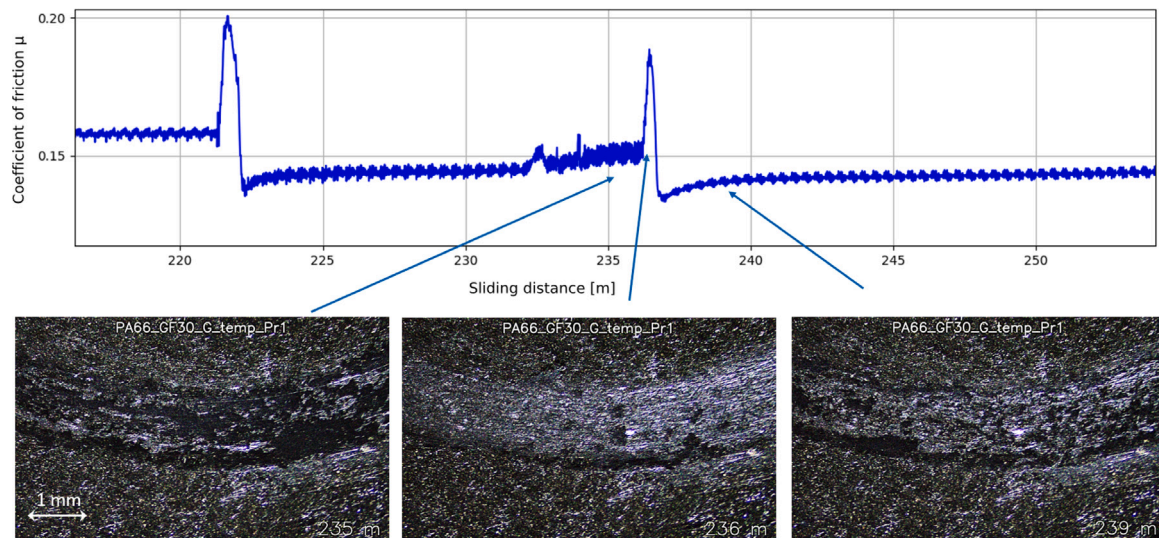


Fig. 9. Coefficient of friction with spikes and microscopic images of the sliding path for different sliding distances with graphite lubrication and tempering (GF30-G-S-F50T), sample “rot_Pr1” out of four. White: Glass fibers. Gray: Unworn matrix PA66. Black, in the sliding path: agglomerated wear particles. Microscopic imaging videos for other sliding distances is published together with this study on the journal’s website.

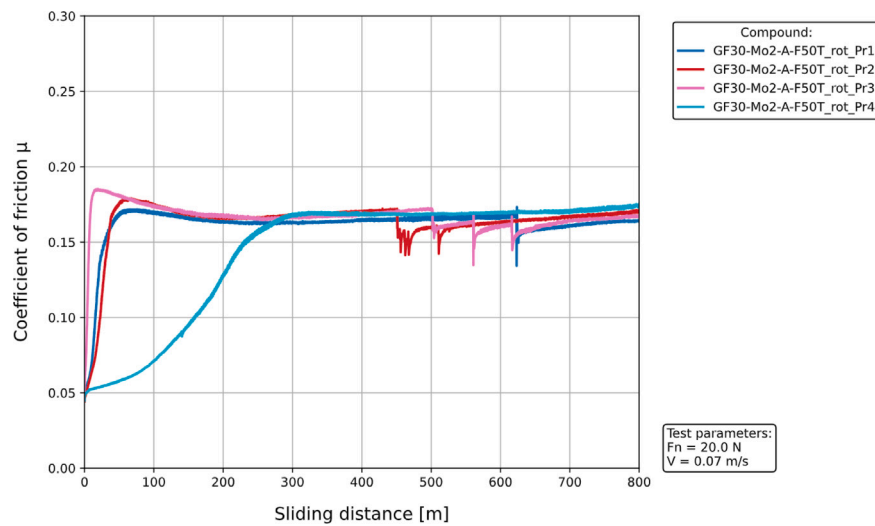


Fig. 10. Coefficient of friction over sliding distance for four samples of GF30-Mo2-S-F50T.

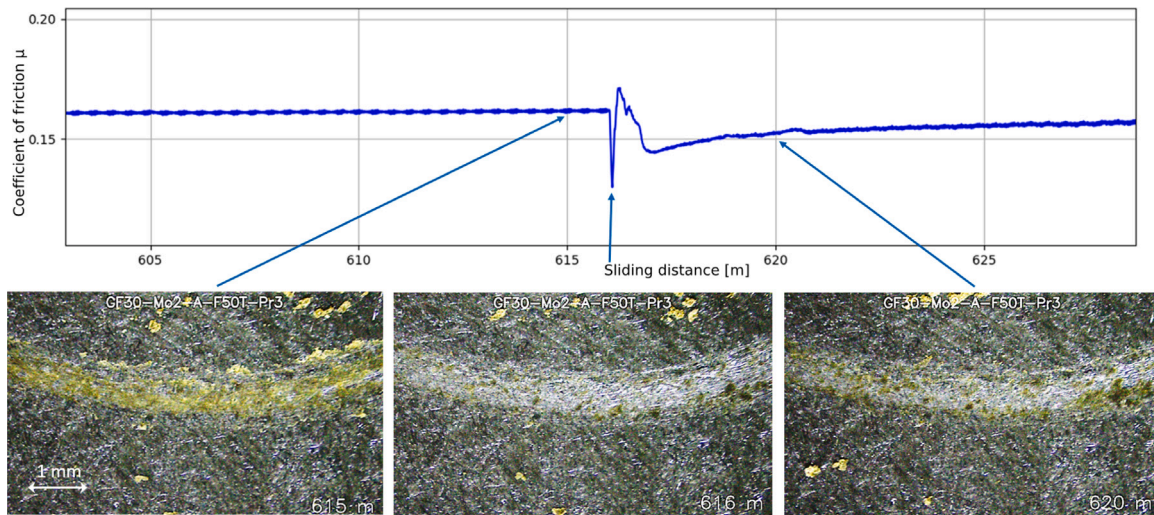


Fig. 11. Coefficient of friction with spikes and microscopic images of the sliding path for different sliding distances with MoS_2 lubrication and tempering (GF30-Mo-A-F50T), sample “rot_Pr3” out of four. White: Glass fibers. Gray: Unworn matrix PA66. Yellow: tribological layer and wear particles. Microscopic imaging videos for other sliding distances is published together with this paper on the journal’s website.

the removal, the coefficient of friction is at a slightly lower level and starts increasing, while the yellow layer is rebuilt slowly. The process is a periodic limit cycle which, in terms of the coefficient of friction, is similar to the graphite-lubricated case, the other lamellar-lubricant system; however, the underlying micromechanical and chemical processes might be fundamentally different.

Similar processes can be observed for other conditioning states of the graphite- and MoS_2 -lubricated samples in this study; however, the effect is less clearly pronounced. This kind of repetitive process is not observed in the samples without solid lubrication, with PTFE, or with polymer lubricant (Fig. 5).

5. Discussion

The high load applied in this study, close to the yield stress in the contact (see Tables 1 and 2), favors the presence of wear particles in the friction interface. Furthermore, it supports a rise in local temperature (although the bulk temperature of the samples in this study remained below 40 °C). The lubrication with PTFE and a polymer lubricant produced low and stable coefficients of friction. However, the two

lamellar lubricants, graphite and MoS_2 , did not show any friction reduction compared to the unlubricated case, independent of the humidity conditioning and heat treatment. Such effects are for graphite (without the presence of glass fibers or PA66) described in [19]. Moreover, the materials with graphite and MoS_2 additives showed quasi-periodic spike events in the coefficient of friction. Microscopic imaging in situ and in operando showed a repeated slow growth and sudden removal of an interface layer.

In the graphite case, this is due to the compaction of worn material into a layer in the frictional path, as microscopic pictures indicate. Triggering events, e.g. due to a rising tangential force, lead to a sudden and complete removal of this layer. A similar process of repetitive layer growth and destruction is known on the surface of high-loaded automotive brake pads, which are composite materials incorporating soft matrix material, abrasive fibers and further substances and therefore are somehow similar to the system under investigation here. The difference is that the nominal contact area between the brake pad and brake disk is rather high compared to that in the ball-on-disk experiment here. Consequently, on brake pads, growth and destruction develop at different surface positions in different phases, leading to a smearing effect. The measured global friction coefficient, which is

a spatial integration, replaces the spikes measured in this study with slower internal dynamics [31,32].

A repeated growth and destruction of an interface layer was similarly found for the compound including MoS₂. The appearance of the yellow layer and wear material, which was not present in the original material, must result from a chemical reaction triggered by the presence of MoS₂.

The appearance of a yellow surface layer and yellow wear debris in the MoS₂-lubricated samples is a reproducible experimental observation that was not present in the original material. Several authors reported the formation of a tribo-chemical oxidation of MoS₂ to molybdenum oxide, in particular yellow MoO₃, which can be assumed here as the underlying mechanism [33–35] here. The observed in-situ and in-operando dynamics in this study showed the repeated removal of MoO₃ from the interface. This is in accordance with others researchers' findings, who report that the MoO₃ layer has weak adhesive strength and can easily be worn off polyamide material [23].

Based on literature reports, a tribo-chemical oxidation of MoS₂ to molybdenum oxides, in particular MoO₃, represents a plausible explanation for this color change under sliding in oxygen-containing environments [23,35].

Humidity and thermal conditioning influence the tribological response of glass fiber reinforced PA66; however, their effect is secondary compared to the dominant role of solid lubrication and surface dynamics under high load. The experimental results indicate that humidity primarily affects the mechanical response of the polymer matrix rather than introducing distinct friction mechanisms. All conditioning treatments that make the PA66 matrix less ductile (drying, tempering) reduce the coefficient of friction, while conditioning treatments that make the matrix more ductile (increased humidity) increase the coefficient of friction.

The results suggest that the lubrication mechanisms of graphite and MoS₂ are suppressed under the severe loading conditions applied in this study. Under reduced contact pressures, where lamellar shear mechanisms are expected to dominate, lower friction coefficients are anticipated, motivating further investigations, e.g. in journal bearing-type contacts.

6. Conclusions and perspectives

The following conclusions and perspectives for next research steps can be drawn from the present study:

1. Solid lubrication strongly determines the friction level of GF-PA66 under the applied severe loading conditions. PTFE and the inferred polymer-based lubricant lead to low and stable coefficients of friction, whereas graphite and MoS₂ do not reduce friction compared to the unlubricated reference.
2. The synchronized in situ and in operando imaging approach reveals that friction evolution is closely linked to surface dynamics. In particular, the graphite- and MoS₂-filled compounds show a repeated build-up and sudden removal of interfacial layers, which directly corresponds to characteristic spikes in the coefficient of friction.
3. For graphite-filled GF-PA66, the observed cyclic behavior is attributed to the compaction of wear debris and graphite particles into a mechanically unstable surface layer. Its sudden breakaway resets the friction level and initiates a new cycle of layer formation.
4. For MoS₂-filled GF-PA66, the appearance of yellow wear debris and surface layers indicates a tribo-chemical reaction during sliding. Based on literature reports, the formation and subsequent removal of molybdenum oxides, in particular MoO₃, is considered a plausible explanation for this observation.

5. Humidity and thermal conditioning influence the tribological response, but their effect is secondary compared to the lubricant-specific interfacial mechanisms. Lower matrix ductility, caused by drying or tempering, tends to reduce friction, whereas increased humidity leads to higher friction under the investigated conditions.
6. The results show that lamellar solid lubricants cannot be assumed to provide low friction in highly loaded GF-PA66 contacts. Future work should therefore investigate lower contact pressures, alternative contact geometries such as journal-bearing-type configurations, and chemical surface analysis techniques such as Raman spectroscopy, XPS, or EDS to verify the composition of the observed tribological layers.

CRediT authorship contribution statement

Matthias Graf: Writing – original draft, Supervision, Resources, Project administration, Funding acquisition, Conceptualization. **Nils Thoben:** Visualization, Validation, Software, Investigation, Formal analysis, Data curation. **Timo Knöner:** Software, Investigation, Formal analysis, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.triboint.2026.112221>.

Data availability

Data will be made available on request.

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